

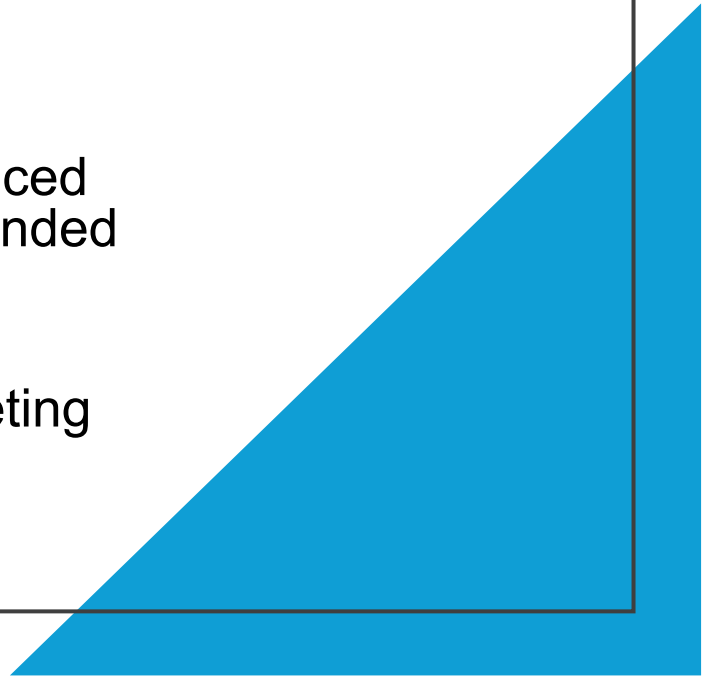


Leonardo Borchert

Mechanical & Aerospace Engineer
Entrepreneur

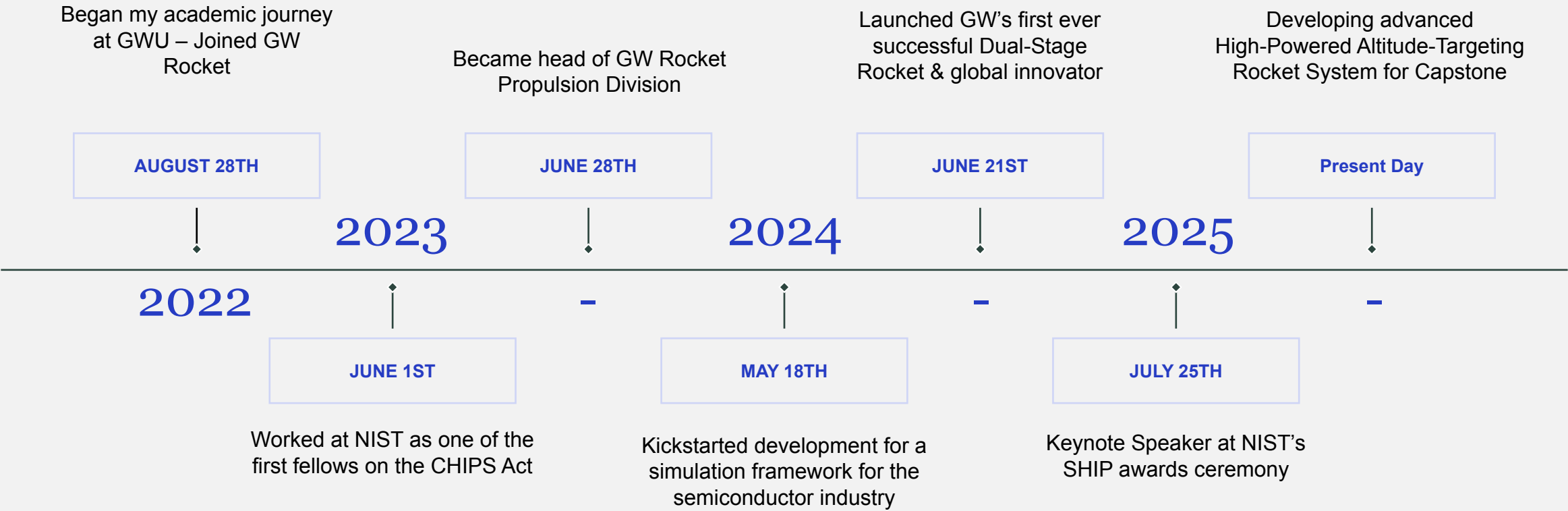
Introductions:

- I am a Mechanical and Aerospace Engineering student at The George Washington University, a three-time NIST SURF research fellow, and the Propulsion Lead of the GW Rocket Team.
- I have had the privilege to work with cutting-edge quantum photonics research, contribute to CHIPS Act R&D, lead advanced propulsion projects, and am planning to build a company grounded in engineering innovation.
- My path has been shaped not by luck, but by preparation meeting opportunity.



College Journey

FOUR YEARS MAKING HISTORY



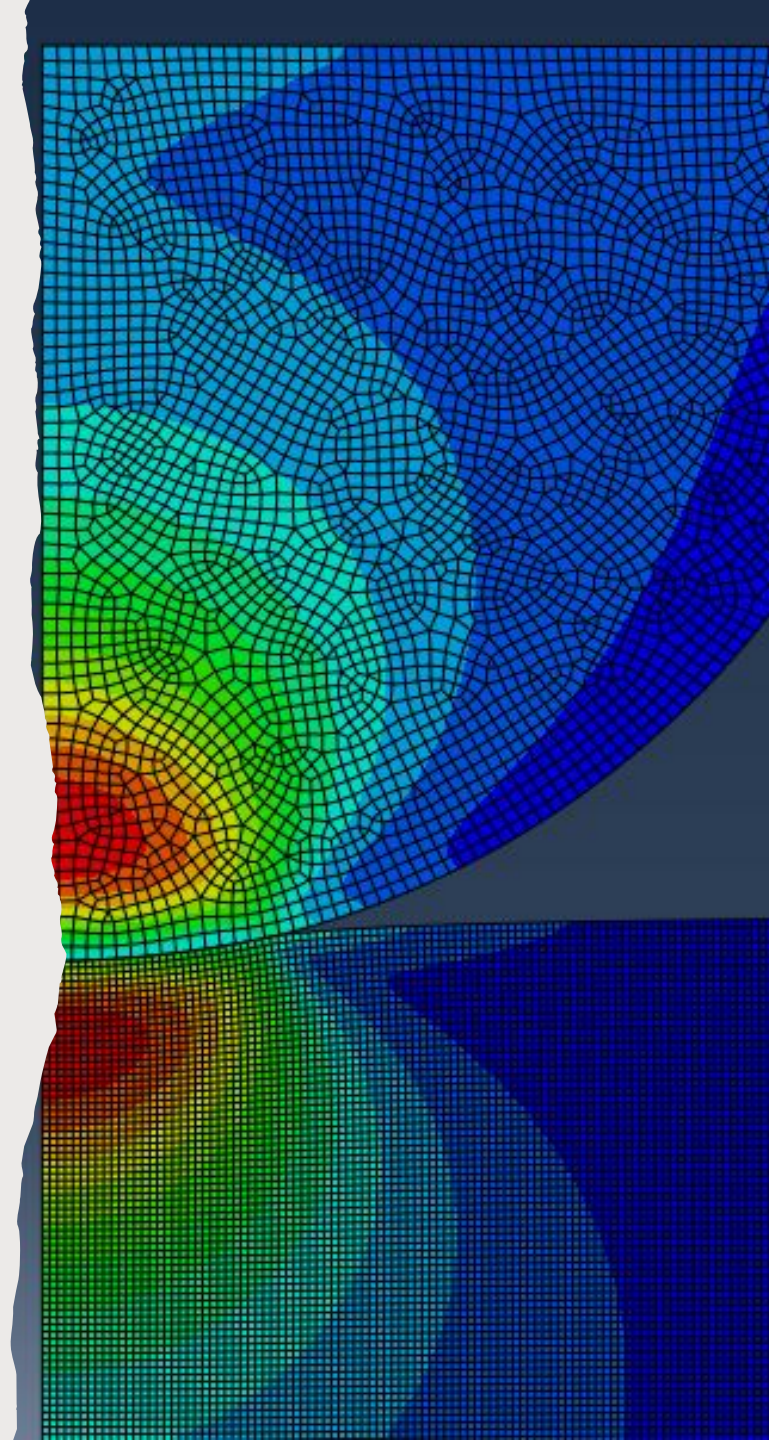
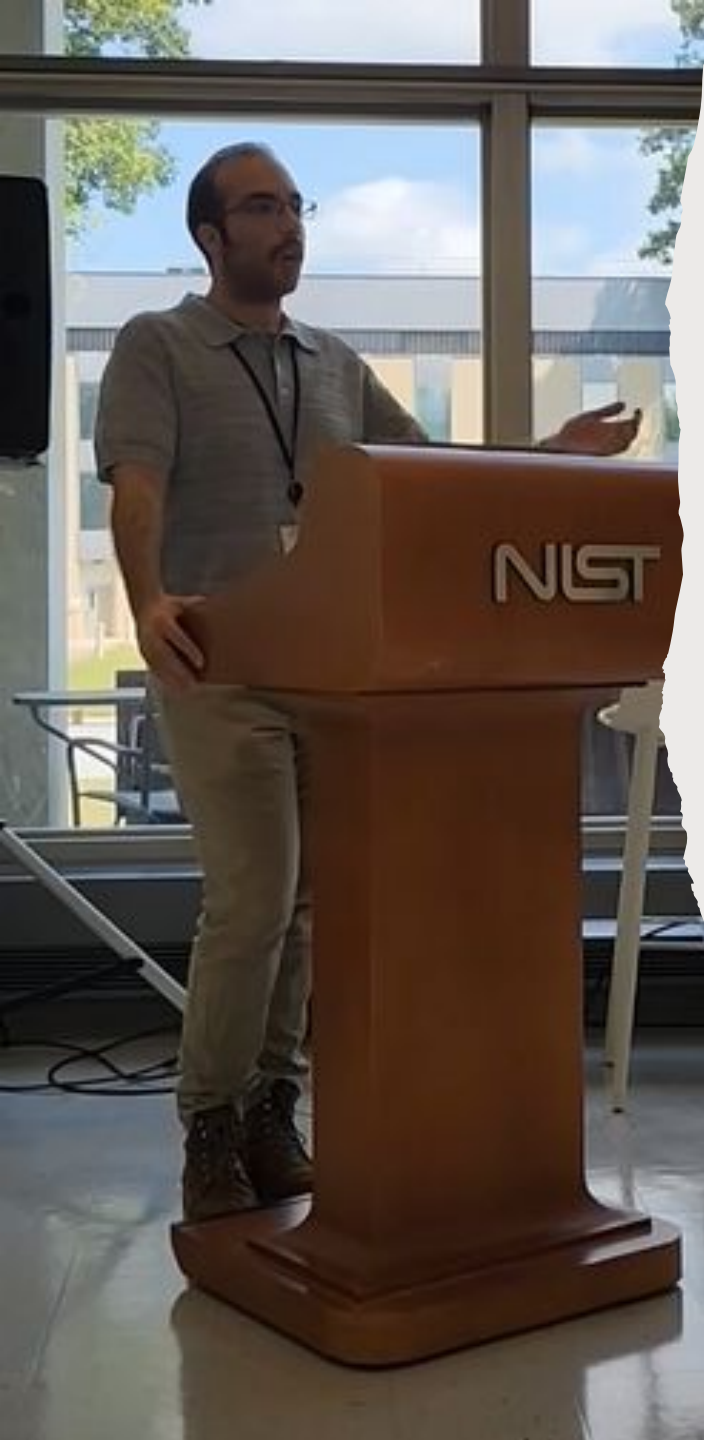
GW ROCKET TEAM

- Promoted to Propulsion Division Lead as a sophomore becoming one of the youngest in leadership
- Led GW's first successful dual-stage solid motor rocket launch at Spaceport America Cup 2024
- Made GW the first university globally to implement live in-flight motor performance analysis
- Positioned GW among 24 universities worldwide selected for innovative rocket technology presentations
- Spearheaded the design and development of custom solid rocket motors and novel motor retention systems
- 4 Secured major sponsorship: Dell Technologies



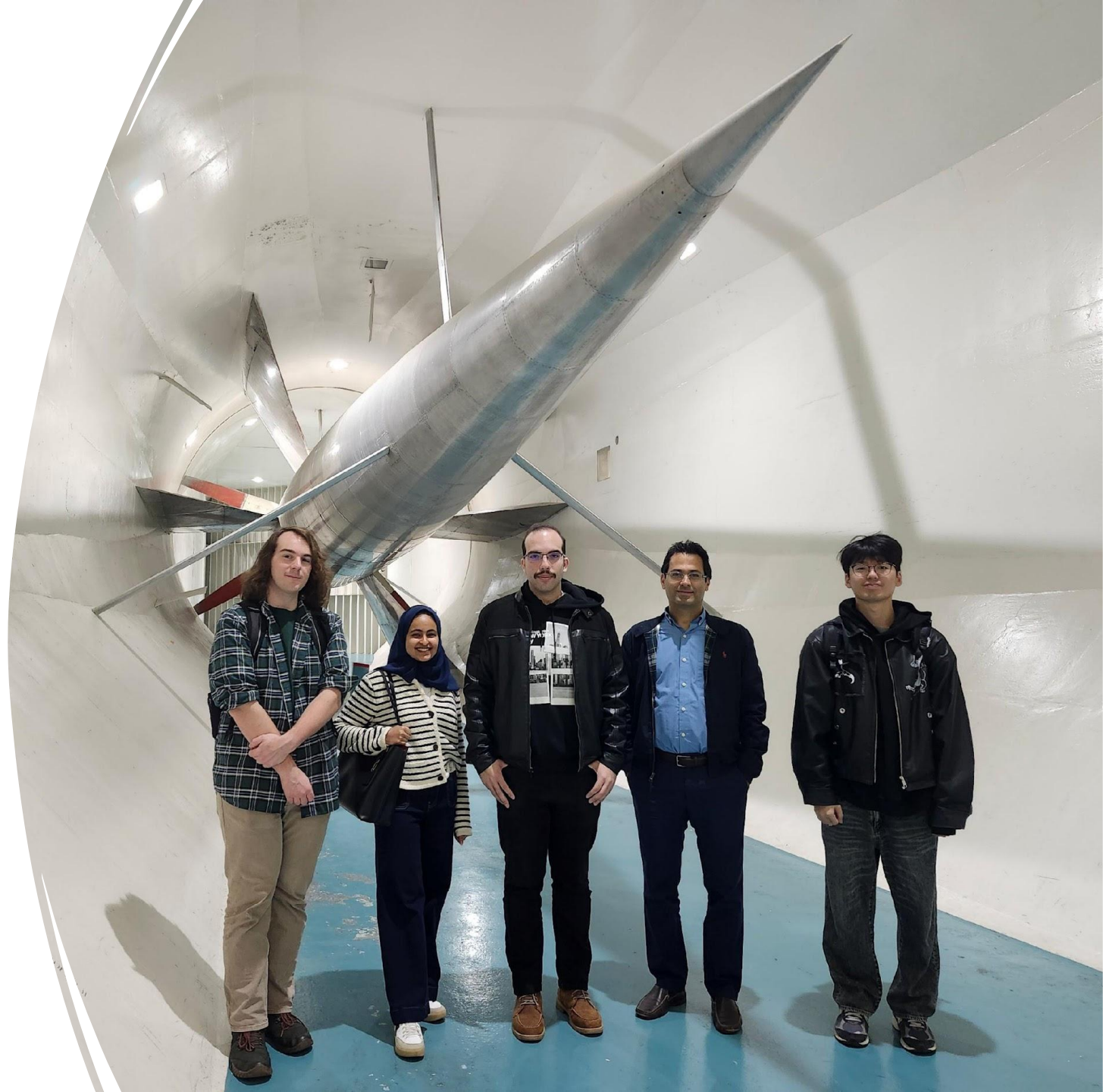
Working at NIST

- Accepted into the **highly selective NIST SURF Program** ($\approx 12\%$ acceptance rate)
- Hired and worked as part of the **CHIPS Act R&D** for semiconductor industry advancement.
- Optimization of nano-calorimeter sensors using finite element analysis.
- Finite element analysis of elastic indentation deformation of layered structures in advanced packaging.
- Measured interfacial bonding strength of quantum photonic devices by indentation-induced blistering.
- Served as a **Judge** at NIST's Summer Highschool Internship Program (SHIP).
- Invited to be **Keynote Speaker** at NIST's SHIP Awards Ceremony by the Deputy Director of Academic Affairs.



Prepared for Capstone

- Developing an **11ft high-powered altitude-targeting rocket system** from concept to launch readiness
- Conducted an engineering tour of the University of Maryland's subsonic wind tunnel to inform aerodynamic decisions
- Expanding into computational fluid dynamics (CFD) to analyze flow behavior, drag characteristics, and flight performance
- Capstone helps serve as a platform to grow teamwork, communication, and engineering leadership skills



A photograph of two hikers on a rocky mountain peak. One hiker in a yellow jacket is standing and reaching down to help another hiker in a red jacket who is sitting on the ground. The background shows a cloudy sky and a sunset or sunrise glow. The image is partially covered by a purple gradient at the bottom.

How can you Achieve Success?

- **Strong communication is non-negotiable**
 - Learn to text clearly, email professionally, and pick up the phone when it counts
- **Work ethic matters more than talent**
 - Show up prepared, follow through, and take pride in what you deliver
- **Be the teammate everyone can rely on**
 - Consistent, supportive, patient, and focused on the group, not just yourself
- **Help others when you can**
 - Your success grows when the people around you grow
- **Engineering is collaborative**
 - The best solutions come from teams that communicate openly and trust each other

Thank You MAE 1001!

Questions?



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